AUTOSAR HIGH-LEVEL DESIGN (HLD) SPECIFICATION

System Component: ADAS Domain Controller Software Architecture

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1. System Overview

Specifies the AUTOSAR Adaptive and Classic mixed platform architecture for the Advanced Driver Assistance System (ADAS) Domain Controller. Handles Level 2+ Automated Emergency Braking (AEB), Adaptive Cruise Control (ACC), and Lane Centering Assist (LCA). ASIL D safety requirements apply.

2. Software Components

Component Name (SWC)	SWC Type	Safety Level	Responsibility
`SWC_CameraPerception`	Adaptive Application	ASIL B	Neural object detection for vehicles, pedestrians, lanes.
`SWC_RadarFusion`	Adaptive Application	ASIL B	Kalman filter fusion of 77GHz front radar and 4 corner radars.
`SWC_TrajectoryPlanner`	Application SWC	ASIL D	Computes collision-free longitudinal/lateral acceleration trajectories.
`SWC_VehicleMotionControl`	Actuator SWC	ASIL D	Translates trajectories into torque requests and steering angle requests.